



Fig. 11: Schematic diagram of a hybrid lithium-air battery

conductivity of $7 \times 10^{-4} \text{ S cm}^{-1}$ at RT. Another important NASICON type electrolyte is a compound of lithium germanium aluminium phosphate (LAGP) that demonstrates a conductivity of $2.4 \times 10^{-4} \text{ S cm}^{-1}$.

For the fabrication of cathodes (air electrodes), carbon materials like carbon fibre, carbon nanotubes (CNTs), graphene nanosheets, reduced graphene oxide, etc are widely employed. This is because they are cost-effective, possess a desirable electronic conductivity, have large surface area and catalyst activities. Moreover, they can be easily mass produced. Further, carbon is prone to form a favourable interface between carbon and solid electrolyte particles by mechanical milling or heat treatment.

The air electrode is made by mixing carbon nanotubes or other suitable carbon materials with ceramic electrolyte (LAGP/LATP) followed by heat treatment. RuO_2 and NiO nanoparticles acting as catalyst are then added. The cells thus produced are able to sustain about 450 cycles (75 days) at ambient air.

Ceramic electrolytes like LAGP/LATP have been closer to commercial applications. However, several crucial issues still remain as obstacles to their practical application. Although the lithium-ion conductivity of the ceramic electrolyte is quite high and the interfacial impedance between the particles in the ceramic sheet can be minimised, the interface contact and interface compatibility of the ceramic and the electrode are challenging.

It is well known that achieving intimate solid-to-solid contact is extremely vital but also extremely difficult to achieve. Poor solid-to-solid contact commonly results in high interface resistance and the formation of lithium dendrites. In addition,

it is very difficult for ceramic sheets to be defect-free, which also results in dendrite formation. Therefore, resolving interface issues is critical for the performance of ceramic electrolyte.

For better physical contact, introducing an alloy layer between the electrolyte and lithium metal or depositing the electrolyte material and the cathode material together are effective. The combination of ceramic electrolyte and polymer electrolyte is also a method to improve the interface engineering. Polymers are more liquid-like than ceramics and solid-liquid contact has inherent advantages over solid-solid contact. However, as the ionic conductivity of polymer electrolyte is lower than ceramic, it is important to find a balance point between the ratio of these two components.

Research work to address these critical issues is being carried out worldwide. It is expected that shortly solid-state lithium-air batteries would be commercially produced that could sustain cycling for at least more than 1000 times at ambient temperature suitable for electric vehicles.

2. Solid polymer electrolyte (SPE) that is defined as solvent-free salt solution in a polymer host material that conducts ions through polymer chains. SPEs are much easier to process and manufacture. They are elastic, giving stability at the interface, and are resistant to volume change during operation.

In general, their ionic conductivity is lower than of ISEs. Despite inferior ionic conductivity ($< 10^{-5} \text{ S}$

cm^{-1}) at RT, and poor electrochemical stability at high voltage, SPEs show high flexibility and low fabrication costs. Batteries with gel type polymer electrolyte made from a compound of polyvinylidene fluoride hexafluoropropylene have a moderate performance of about 180 cycles. A hybrid solid-state electrolyte using polymer and inorganic solid electrolytes is expected to give better performance.

3. Quasi solid-state electrolyte (QSSE) is a composite compound consisting of a liquid electrolyte that acts as a route to conduction of ions and a solid matrix that gives mechanical strength. At present, ceramics, LAGP, and polyethelene oxide (PEO) polymer electrolytes are employed as solid-state electrolytes. In terms of chemical and electrochemical stability, LATP and LAGP may be the most promising electrolyte.

Hybrid electrolyte battery

In this design both aqueous and aprotic electrolytes are used to have the advantages of both aqueous and aprotic battery design, as shown in Fig. 11. A lithium conducting membrane made of ceramic, glass, or polymer separates the two types of electrolytes in the cell.

The lithium anode is coated with lithium-conducting ceramic to protect the lithium metal from violent reaction with water. Li^+ ion transfer between the liquid and solid phase is kinetically sluggish and consequently Li^+ ions mainly migrate through the maze of the liquid electrolyte due to its higher conductivity.

It is important to mention that often the liquid components compromise the thermal stability and the safety of the battery causing explosion. They have a moderate performance of around 100 cycles. Hence, they are not suitable for electric vehicles.

Architecture of carbon cathode

In the near future it will be possible to manufacture commercial lithium-